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Jurassic magnetostratigraphy, 3. Bathonian-Bajocian of Carcabuey, Sierra Harana and Campillo de Arenas (Subbetic Cordillera, southern Spain)

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Four sections in Majocian-Bathonian (Middle Jurassic) pelagic limestone with standard ammonite zonation have yielded magnetic polarity sequences. Magnetic directions in these red to white limestones were obtained by thermal demagnetization and were stable from about 300 °C to in excess of 450 °C. The polarity patterns indicate that the majority of the Bajocian and Bathonian is characterized by quite frequent reversals of the magnetic field. Lengthy periods of constant polarity, particularly constant normal polarity, were not observed. The average frequency of reversals is about 6 per ammonite zone, which roughly may be interpreted as a frequency of a reversal every 260,000 years, a rate comparable to that of the Miocene-Pliocene. Paleolatitudes of these sites (25–28°) are about 10° south of their present positions; variable clockwise block rotations within the Subbectic region have rotated these sites relative to stable Iberia.

1. Introduction

The geomagnetic polarity time scale for the Late Jurassic and Early Cretaceous has been moderately well established by the correlation of magnetostratigraphic sequences in sedimentary rocks with the "M-sequence" of marine magnetic anomalies. The Oxfordian-Kimmeridgian boundary falls approximately at chron M25 [1–3], the oldest magnetic anomaly which can be correlated between the Pacific and Atlantic Oceans.

Linear magnetic anomalies older than anomaly M25 are of low amplitude and cannot be correlated between ocean basins. This pre-Kimmeridgian oceanic crust comprises a magnetic "quiet zone", originally thought to have been formed during a period of constant normal polarity, similar to that of the Late Cretaceous [4,5]. In the Atlantic, the "outer quiet zone" (seaward of the Blake Spur anomaly) has broad low amplitude linear magnetic anomalies, designated "M26–M28" [6], and dated by the drilling of DSDP Site 534 as Callovian through Oxfordian [7]; these anomalies may be wholly or partly caused by topography [8]. The "inner quiet zone" (landward of the Blake Spur anomaly) has been interpreted to have formed during the initiation of seafloor spreading in the Late Bathonian [9].

In the Pacific, the pre-M25 magnetic anomaly pattern has been identified and correlated between different spreading centers [10–13]; if the rate of seafloor spreading remained constant, then the projected age of the Pacific "M29" should be Early Oxfordian, and "M38" should be approximately Late Bathonian [14]. The magnetostratigraphic records of Callovian and Oxfordian sedimentary sections do not correlate very well with these pre-M25 magnetic anomaly patterns, and suggest that there are short-duration polarity intervals not resolved in the seafloor anomaly pattern [14,15]. The combined marine anomaly patterns and sedimentary magnetostratigraphic records suggest that the frequency of reversals in the Late Bathonian through Oxfordian was an average of about 5 per million years, similar to the reversal frequency of the Late Tertiary.

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There is no oceanic crust, hence no marine magnetic anomaly, known to be older than Late Bathonian. It is possible that seafloor as old as Bajocian exists in the Pacific, but attempts to determine this by drilling have been unsuccessful [16,17]. It is necessary, therefore, to derive the magnetic polarity time scale for the Middle Bathonian and older stages entirely from the magnetostratigraphic records of sedimentary rock sequences.

There have been few paleomagnetic investigations of the Bajocian or Bathonian stages. Catalano et al. [18] reported a very high frequency of reversals from two Bajocian-Bathonian sections of red to pink nodular pelagic limestone in western Sicily. They observed 20 and 38 polarity zones respectively in the two sections (although most of the polarity zones were represented by single samples), but biostratigraphic control is lacking. Ogg and Channell examined poorly dated uppermost Bajocian and Lower Bathonian red nodular pelagic limestone in the Southern Alps [19,20]. They obtained mixed polarity with frequent reversals in portions of some sequences, but were unable to correlate between sections. Mixed polarity in Bajocian-Bathonian red pelagic limestone in Hungary was reported, but in abstract only [21]. Hijab and Tarling [22] studied Bathonian limestone in Iraq and obtained a mixed polarity sequence that they called an interval of "unstable geomagnetic polarity". Thus by most indications, the Bajocian-Bathonian appears to be an interval of mixed, and probably rapidly changing, polarity.

In this report, the magnetic polarity of four, well-dated, overlapping sections is presented. These sections are primarily red to light gray pelagic limestone and marl, encompassing most of the Bajocian and Bathonian. The Late Bajocian and most of the Bathonian are characterized by a high frequency of reversals, such that detailed correlation between sections is not possible on the basis of magnetostratigraphy alone. This high reversal frequency occurs just prior to the time of the marine Jurassic "quiet zone", and a similar high frequency is also found in the Oxfordian, the young end of the quiet zone. These two occurrences of high frequency reversals bracketing the quiet zone suggest that rapid reversals may contribute to the origin of the low amplitudes of anomalies of quiet zone crust.

2. Geologic setting and stratigraphy

The Subbetic Cordillera is a 40–80 km wide belt of Jurassic and Cretaceous pelagic sediments extending 500 km across southern Spain (Fig. 1). Pelagic sedimentation began in the early Pliensbachian stage of the Early Jurassic and continued until tectonic shortening began in the Eocene [23].

Deposition during the Jurassic occurred on two submarine highs which were separated by a median trough. This general topography and the irregular relief upon the highs led to considerable lateral variation in the Jurassic facies from late Early Jurassic (Pliensbachian) onwards [24,25]. In general the Bajocian-Bathonian strata deposited on the highs are represented by light gray to pinkish biomicritic limestone, nodular limestone and red marl, whereas the basinal facies are thicker sequences of interbedded darker gray siliceous micritic limestone and marl. Three of the four sections described here were in the facies deposited on the submarine highs, for two reasons: (1) the lighter gray, and especially the pinkish, pelagic limestone displays magnetic properties more suitable for paleomagnetic studies than its dark gray, more reduced, basinal counterpart, and (2) the submarine highs were above the regional aragonite compensation depth (ACD), and thereby generally preserve a more detailed ammonite biostratigraphic record. However, the sections on the highs display numerous hiatuses. The latest Bathonian, Callovian and early Oxfordian of these occur in extremely condensed facies and can be recognized in only a few localities.

Compression between Africa and Iberia in the Tertiary produced northward thrusting, folding, and considerable lateral shear with associated rotation of local blocks [23]. As a result, the paleomagnetic directions from our sections, located in different structural blocks, have little significance for the net motion of the Iberian plate.

Detailed ammonite biostratigraphy in the Bajocian-Bathonian of the Subbetic region has been established by Sandoval [25–27], and his zonation has been followed in this study. The zonation utilizes standard zones with three exceptions: (1) the "*S. Sowerbyi* zone" has since been subdivided into the Discites and Laevinscula standard zones (see [28]); (2) the "*O. Aspidoides* zone" has been

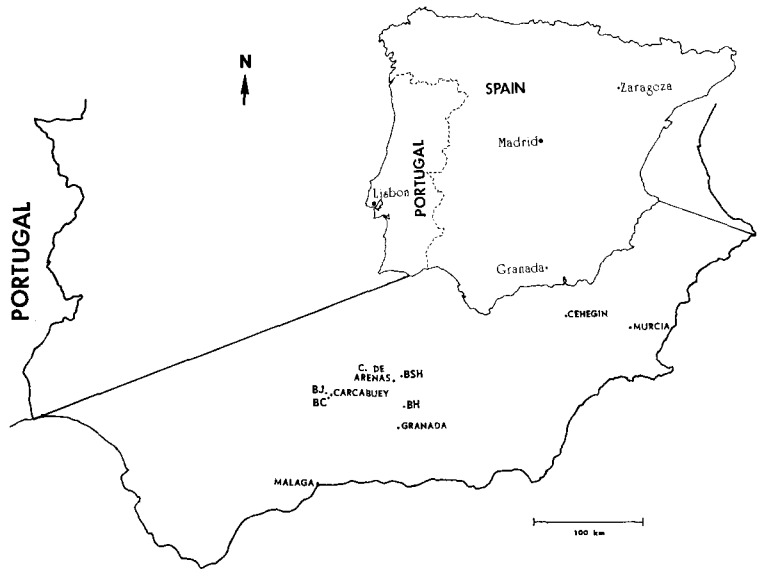


Fig. 1. Map showing the location of the four main sequences discussed.

replaced by the *O. Orbis* Standard Zone [29]; and (3) the Spanish biozones, "*Bullatimorphites costatus*" and "*Tulites (Rugiferites) sofanus*", are used to help subdivide the Middle Bathonian which does not have standard zones. A convention for the names of Jurassic standard zones is not followed worldwide, therefore for clarity the initial of the genus will be used with the species name. Westermann [30,31] recently has assessed Jurassic biostratigraphic information in detail and has estimated that the average duration of Jurassic ammonite zones is about 1 m.y.

The placement of ammonite zone boundaries within the sampled outcrops has several minor uncertainties. First, zone boundaries, defined by the occurrence of guide ammonites, were usually assignments based on the faunal content of a specific stratigraphic bed. Since the average thickness of a bed is about one half meter, there is an inherent uncertainty of ± 50 cm. Second, because the collection of a particular fossil may not be its actual first occurrence, the actual zone boundaries may be slightly lower or higher. Third, Late Bajocian zones are less precise than others, because the guide fossils are rare. A fourth, more significant source of uncertainty is the duration of hiatuses in the sections. Hardgrounds, erosion surfaces and sudden changes of lithologic character are present throughout the outcrops. Although these hiatuses rarely comprise more than a small portion of an ammonite zone, they make magnetic polarity correlation difficult or impossible in intervals where the reversal frequency is high (several polarity zone per ammonite zone) and sedimentation rate low. This is particularly true for the *S. Subfurcatum* Standard Zone of the early Late Bajocian and the condensed *Tulites (Rugiferites) sofanus–Bullatimorphites costatus* biozones of the Middle Bathonian.

Eight paleomagnetic sections were sampled during this study. Of these, only four yielded useful paleomagnetic data. Their locations are shown in Fig. 1, and their stratigraphy is summarized below. The sections span the Early Bajocian through the Middle Bathonian. For magnetostratigraphic purposes, the Upper Bathonian is essentially absent in all sections; only strata comprising a portion of a single ammonite zone are present in some sections.

The first section, "Carcabuey" (BJ), is a roadcut located 1 km west of Carcabuey (37.5° N, 3.3° W) and 80 km WNW of Granada. This outcrop is section "JC4" (Puerto Escaño) of Sandoval [25]. The beds dip 45° toward 125°. The section is 33 m thick and consists of the entire Bajocian and Early to Middle Bathonian. The lower part is light gray micritic limestone, and the upper part is interbedded pink biomicritic limestone, pink nod-

ular limestone and red marl. A hardground overlies the Middle Bathonian; above it is red marl of Oxfordian age.

Approximately 2.5 km southwest of the Carcabuey section is a stream valley exposure of the same strata, our "La Fuente" section (BC) and Sandoval's [25] section JC6 (Cañada del Hornillo Este). The uppermost Lower Bajocian through lowest Upper Bathonian (10.6 m) were sampled. The beds dip 11° toward 110°. Compared to the nearby Carcabuey section, the La Fuente section is less marly and has a more pinkish coloration. The La Fuente outcrop lies immediately below our Kimmeridgian-Tithonian section [1].

The third section, "Sierra Harana" (BH), is located about 20 km NNE of Granada (37.2° N, 3.7° W) on the southern flank of the Sierra Harana range (section JA5, Cortijo de Moralejo, of Sandoval [25]). The beds dip 25° toward 195°, and are in a different structural block than the Carcabuey sections. The lowermost 2.2 m, within the upper portion of medium-bedded gray limestones, are barren of ammonites, but are thought to be mid-Bajocian in age. This is followed by 10.5 m of pinkish-gray thin-bedded nodular limestone, which extends from the Late Bajocian through the first ammonite standard zone, *O. Aspidoides*, of the Late Bathonian, of which the first subzone is absent. The contact with overlying red Oxfordian marl is a hardground. This section is one of the most complete Bathonian sequences known in the Subbetics.

The fourth section, "Campillo de Arenas" (BSH), is a stream valley 2.5 km east of the town of Campillo de Arenas and approximately 40 km north of Granada (37.5° N, 3.5° W), section JAC3 (Barranco de la Jarropa) of Sandoval [25]. This section is in the basinal facies of interbedded gray marl and marly biomicritic siliceous limestone. The Lower Bajocian, which Sandoval [25] has dated to the subzone level, is about 65 m thick and has an average dip of 50° toward 300°. Sampling included the uppermost part of the upper ammonite zone of the Aalenian stage through the lower half of the *S. Sowerbyi* zone, and the upper half of the *O. Sauzi* Standard Zone into the *S. Humpriesianum* Standard Zone of the late Early Bajocian. Above this level, the beds rapidly become siliceous and ammonites are no longer preserved.

Four additional sections were sampled, but the paleomagnetic results in these were unreliable. The fifth section, the Upper Bajocian through Middle Bathonian section BCH, was Sandoval's section JCE1, about 5.5 km SSW of the town of Cehegin, or about 200 km ENE of Granada and 55 km WNW of Murcia (38.1° N, 1.8° W). A sixth section, BA, was near Zaragoza, at the small town of Belchite (41.3° N, 0.8° W). Sampling encompassed the entire Bajocian and included reconnaissance through the Bathonian. In addition, we sampled two sequences of Bathonian and Bajocian rocks located on the western coast of Portugal. At Cap do Mondego on the Atlantic coast (40.2° N, 8.9° W), samples were taken in the Upper Aalenian and Lower Bajocian. Along the road between Figuero do Foz and Aveiro, near the town of Brehna (40.2° N, 8.8° W), samples were taken in the Bathonian, Aalenian and Carixian.

3. Laboratory procedure

The sections were sampled during three field seasons. The sampling program began with 20 samples per ammonite zone in sections BJ and BH, but proved to be far too sparse for the high frequency of reversals observed at BJ. Thus, the sampling density at that site was doubled in the second field season, and subsequent sites BC, BSH, BCH, and BA were sampled at the higher density. (BH was not resampled at the higher density).

All measurements were made on a two-axis ScT cryogenic magnetometer with on-line computer data analysis. The noise level is 2×10^{-11} A m^2, or 2×10^{-8} emu, and could be reduced below 5×10^{-12} A m^2 by using multiple readings at each measurement position. A steel shielded room housed the magnetometer and the non-inductively-wound furnace for thermal demagnetization. Each sample was measured in eight orientations (90° rotations around the *Z*-axis in upright and inverted sets) using multiple readings at each orientation for weak samples.

Pilot samples were analyzed to determine the magnetic characteristics of the various lithologies in each section; progressive thermal demagnetization at 30–50° C increments, from 150° to 650° C was performed. AF demagnetization was not implemented as, in the past, thermal demagnetization has proven to be superior in separating the

remanence components in pelagic limestones such as these (e.g. [1]). Based on the pilot samples, thermal demagnetization programs were selected separately for each section. Generally samples were treated with at least five thermal steps. Magnetic polarity and the characteristic direction of remanent magnetization for each sample was determined from either the stable end points, least squares analysis or in some cases, from the behavior displayed during demagnetization, using vector diagrams and equal area plots.

4. Remanent magnetization

The two Carcabuey sections, BJ and BC, and the Sierra Harana (BH) site yielded easily interpretable data. Locality BSH displayed directions extensively overprinted by the present day field direction, but retained enough of a characteristic remanence to suggest the polarity pattern. Section BCH was even more extensively overprinted, and very little polarity information could be obtained. Section BA was weakly magnetized and also was appreciably overprinted by the present field. The Portuguese sections had extremely weak remanences, and were primarily magnetized in the present field direction. Pilot AF demagnetization of the Portuguese samples did not move the sample directions from the present field direction, and pilot thermal demagnetization did not yield a coherent group of directions. Hence, no data were obtained from the Portuguese sections, and only a minimal number of characteristic directions from BA and BCH.

4.1. Carcabuey sections BJ and BC

The natural remanent magnetization (NRM) directions of these two sections, which are separated by only a few kilometers, are shown in Fig. 2 (top). The NRM directions, although lying in proximity to the present and the axial field directions, are statistically distinct from them. The directions are entirely of normal polarity. Intensities vary, but average 1.2×10^{-3} A/m for BJ, and 4.7×10^{-3} A/m for BC. The higher intensities at section BC might be related to the greater amount of red coloration at this locality.

Thermal demagnetization has a fairly dramatic effect on these samples. Approximately 50% of the samples reverse their polarity in response to thermal treatment. Fig. 3 displays two typical samples. By 200 °C, intensities generally have dropped sharply. At 250–300 °C, intensities have declined to 10–20% of the NRM intensity. Usually by 300 °C, or occasionally 350 °C, polarity definition is clear. Univectorial behavior and stable end points are usually exhibited from 300 ° or 350 °C through 450 °C, after which directions sometimes deviate from linear behavior in an apparently random manner, however, the general directions are retained through heating to 500 ° or 550 °C. Above 550 °C, directions were generally random, and intensities were quite weak.

Except for the difference in NRM intensities, which was retained through demagnetization, the two sites behaved in exactly the same manner, as would be expected from their close proximity. Remanent intensities at 450 °C average 8.4×10^{-5} and 1.3×10^{-4} A/m, for BJ and BC respectively. Dispersion was a minimum in both sections at 450 °C, and the sample directions form tightly clustered antipodal groups as shown in the lower half of Fig. 2.

The main objective of this study was polarity determination; however, least squares analysis [32] of the BC section was performed to determine how closely the 450 °C directions approximated those of the linear segments of the demagnetization paths. A comparison of mean direction and statistics of the 450 °C demagnetization step to the directions determined by least squares analysis showed no significant difference (Table 1).

The mean directions from the BJ and BC outcrops are statistically identical after their respective structural corrections. The structural attitudes are considerably different, and the agreement of mean directions demonstrates the pre-folding (pre-Tertiary) origin of the magnetization.

4.2. Section BH

NRM directions at this site are clustered around the present/axial field directions and form a streak of those directions toward the west, as shown in Fig. 4. All NRM directions are of apparent normal polarity. The path of smearing in this distribution is the same as that of the demagnetization paths of samples showing reversed polarity upon thermal demagnetization (see Fig. 5). NRM intensities are slightly higher than those of the BJ/BC localities, averaging 3.0×10^{-3} A/m.

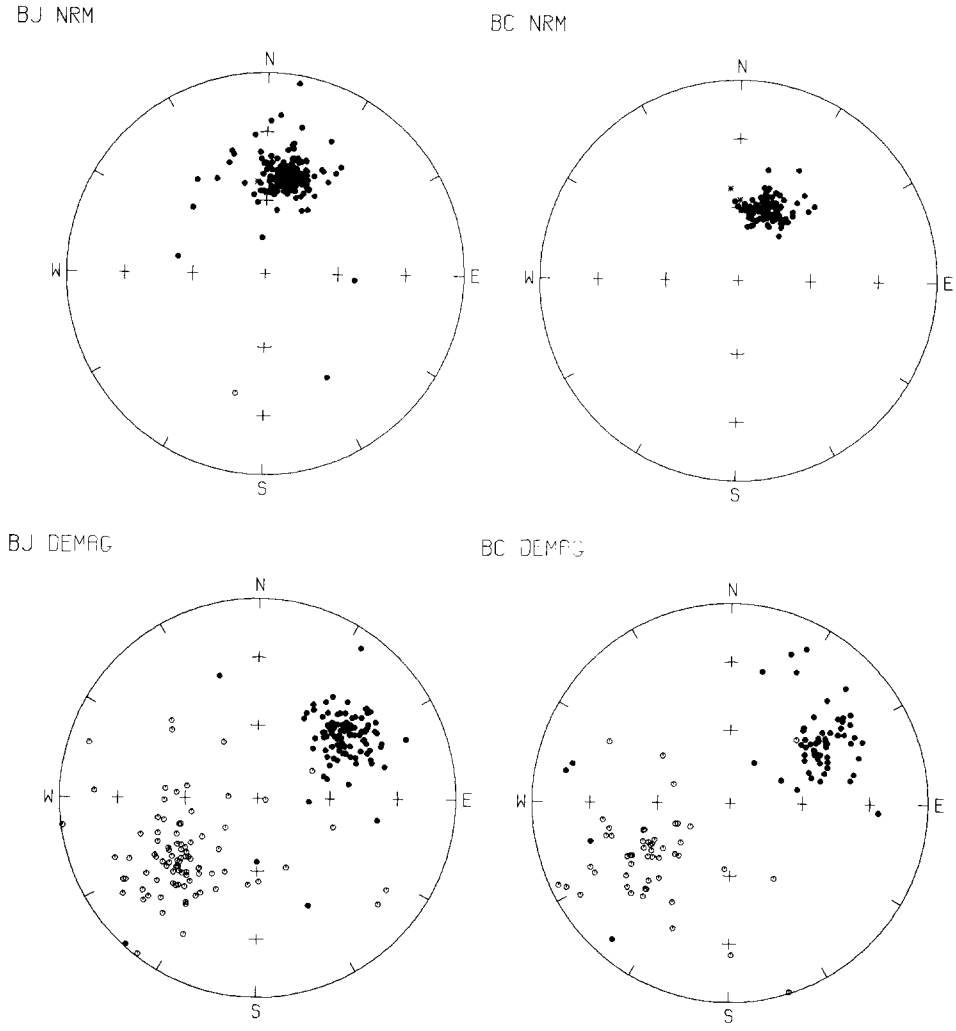


Fig. 2. NRM and thermally demagnetized directions of localities BJ and BC. Equal area projection, solid (open) symbols represent lower (upper) hemisphere directions. Asterisks represent the axial and present field directions at the site. NRM directions are before correction for bedding structure, while demagnetized directions are after correction.

Thermal demagnetization elicited much the same response as observed at the BJ/BC localities. Fig. 5 shows two samples that typify the response of the majority of samples from this outcrop. Slightly more than 50% of the samples displayed reversed polarity upon demagnetization. The reversed or normal directions are generally reached by 350 °C, in contrast to the 300 °C of the other localities. Again, the best grouping of directions, and the most frequent occurrence of stable end points is at 450 °C. The directions are commonly retained through the 500 °C step, after which directions begin to scatter widely. Remanent intensities at 450 °C were 2.1×10^{-4} A/m.

Directions are not as well grouped at BH after demagnetization as they were at the other localities (Fig. 4). The normal and reversed clusters are not antipodal. Both observations indicate the retention of some overprinting magnetization. Supporting these observations is the slightly higher unblocking temperature (350 °C vs. 300 °C) at this site, and the greater overprint on the NRM directions. A probable explanation of this difference is a difference in outcrop settings. BH is a rolling hillside outcrop, a nearly horizontal exposure which probably has been subjected to more

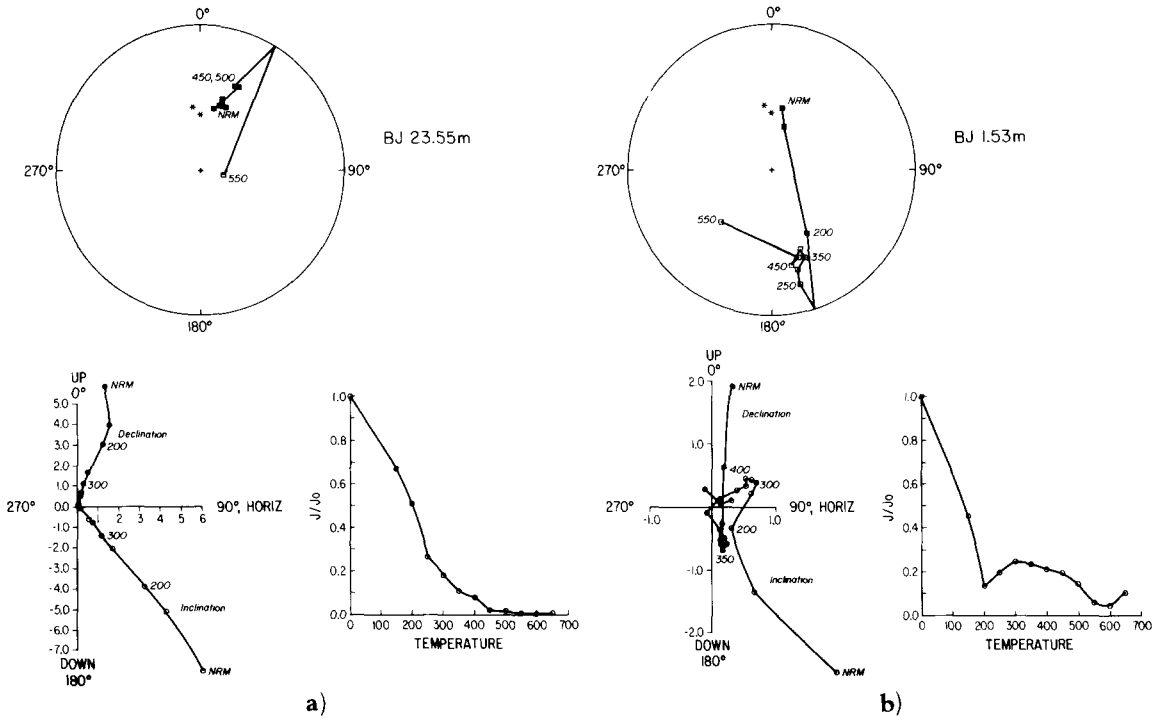


Fig. 3. Thermal demagnetization response of two typical samples from site BJ. Stereonet projection and symbols are as in Fig. 2. Vector diagrams utilize the method of Kirschvink [32], in which the inclination is plotted in the plane of the declination. Intensities are in units of 10^{-4} A/m (10^{-7} emu/cm^3).

weathering processes than the road cut and vertical stream exposures of BJ and BC.

A mean direction was computed from the 450 °C demagnetization step. Despite the somewhat greater overprinting at BH, the statistics of its mean direction are similar to those of BJ/BC (Table 1). The significantly different mean direction of BH relative to BJ/BC reflects the location of the site in a different structural block.

4.3. *Section BSH*

This basinal facies locality in the lower Bajo-

TABLE 1

Thermally demagnetized mean directions

Locality	$D(°)$	$I(°)$	N	R	K	$\alpha_{95}(°)$	Long. pole	Lat. pole	d_p	d_m	Paleolat.
BJ	53.8	47.0	196	181.7	13.6	2.8	81.0 °E	44.4 °N	2.4	3.7	28.2
BC [a]	57.2	43.3	113	103.0	11.2	4.2	82.7 °E	40.4 °N	3.2	5.2	25.2
BC(LSQR) [a]	59.4	42.9	112	102.9	12.1	4.0	81.8 °E	38.6 °N	3.1	5.0	24.9
BH	75.8	46.6	107	93.8	8.0	5.2	80.6 °E	27.1 °N	4.3	6.6	27.8
(BSH	353.5	32.6	115	87.3	4.1	7.5)					
Kimmeridgian and Tithonian poles from BC [1]:											
KC	23.0	35.5	57		56.3	2.5	121.0 °E	63.1 °N	1.7	2.9	
TC	30.4	41.6	48		160.1	1.6	104.5 °E	60.7 °N	1.2	2.0	

N = number of samples; R = resultant vector length; K = dispersion parameter; α_{95} = 95% circle of confidence; d_p, d_m = ovals of 95% confidence of pole; Paleolat. = paleolatitude of site.

[a] BC = mean of 450 °C step of thermal demagnetization; BC(LSQR) = mean of least squares analyses of the linear portions of each sample's demagnetization path.

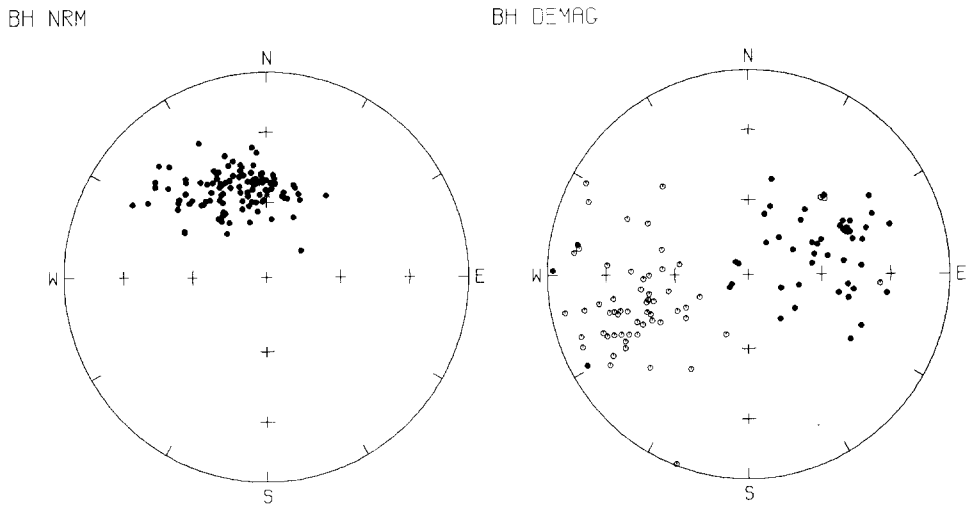


Fig. 4. NRM and demagnetized directions of site BH (equal area projection, symbols as in Fig. 2). NRM directions are before, and demagnetized directions are after, correction for bedding structure.

cian proved to have a poorer magnetic record than the comparable age portion of the BJ section and the other localities. Intensities are an order of magnitude weaker (average = 1.4×10^{-4} A/m), and NRM directions are more scattered than at the other localities (Fig. 6a). Greater variation in NRM intensities is also a characteristic.

In this sample suite, remanence directions were

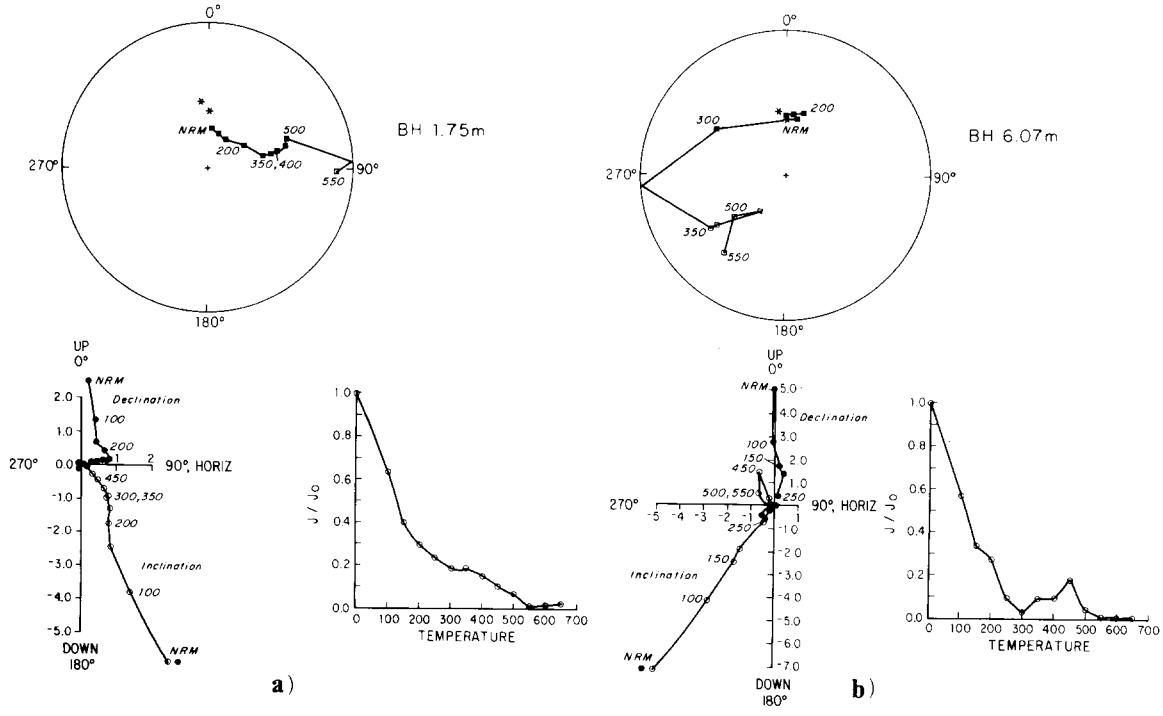


Fig. 5. Response to thermal demagnetization of two typical samples from the BH locality. Stereographic and vector diagrams are as in Figs. 2 and 3. Intensities in units of 10^{-4} A/m (10^{-7} emu/cm^3).

not stable beyond 300 °C during thermal demagnetization. Thermal demagnetization to 300 °C removed 80% of the remanent intensity (average = 3×10^{-4} A/m), and coherent magnetization generally was not observed beyond that temperature. About a third (38%) of the samples display stable end points during demagnetization. Most of these are of normal polarity, which is consistent with the dominantly normal polarity indicated by the BJ section for much of this time. The characteristic magnetization of the remaining two-thirds of the samples was not isolated by 300 °C, and continued demagnetization was not helpful. At 300 °C, sample directions are quite scattered (Fig. 6b), and vector demagnetization diagrams suggest that many samples still contain multicomponent magnetization.

However, a large number of these multicomponent samples exhibit directional trends during demagnetization that follow those of the normal or reversed samples (Fig. 6b). Although stable end points are not reached, opposite polarity demagnetization behaviors are clearly observed in these incompletely cleaned samples. Directions of the apparent reversed samples generally fall much shorter of the true direction than do the normal samples, a consequence of the normal polarity of the overprint. Since the characteristic directions could not be isolated, they cannot be proved to be a Jurassic (Bajocian) magnetization. Nevertheless, for the sake of curiosity, a tentative polarity sequence was constructed, using those samples exhibiting stable end points and the inferred polarity of the multicomponent samples. Any ambiguous

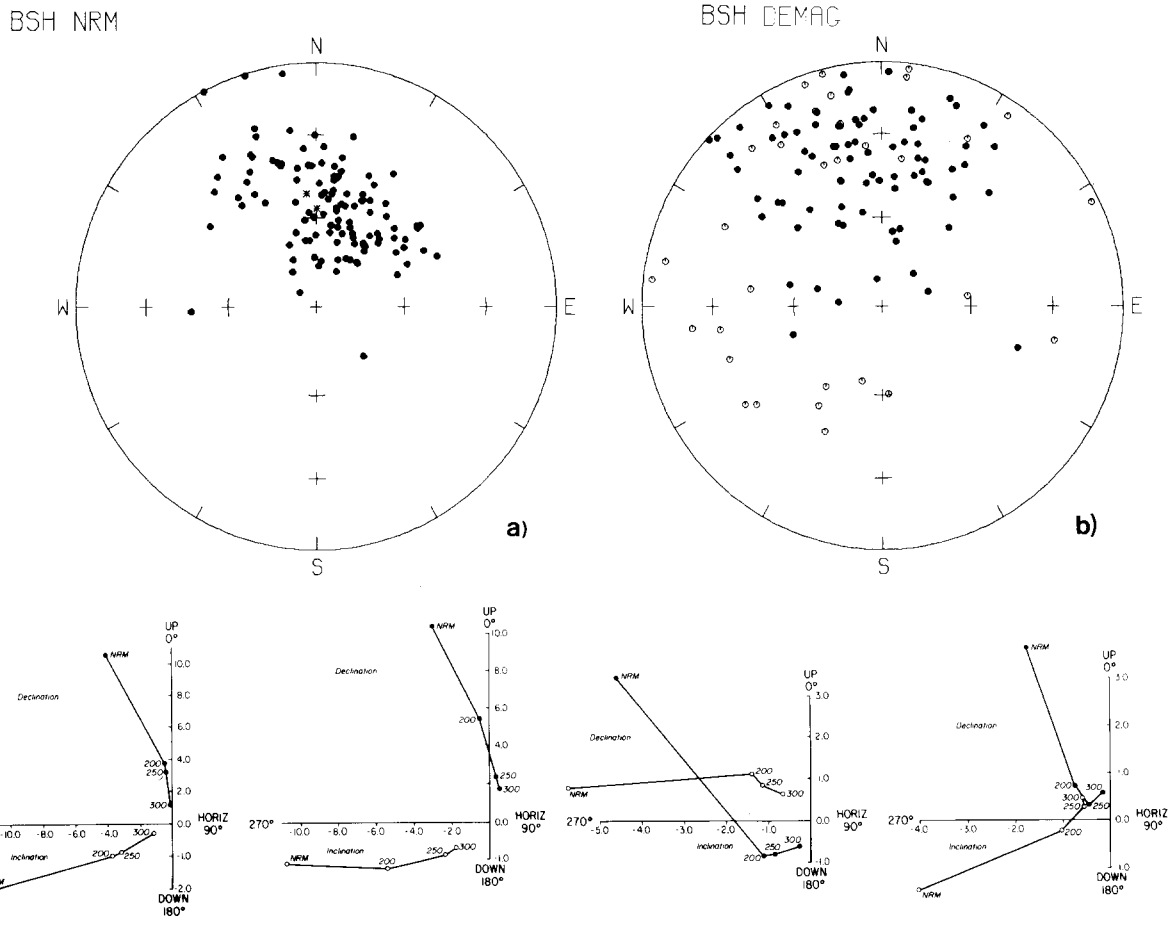


Fig. 6. NRM and demagnetized directions from the BSH locality, and vector diagrams of normal and reversed samples exhibiting both characteristic and multicomponent magnetization. Stereographic and vector diagrams are as in Figs. 2 and 3. Intensities are in 10^{-5} A/m (10^{-8} emu/cm^3).

or present field directions were omitted from consideration. No reference was made to the polarity sequence of BJ during this exercise, and it was therefore surprising that the resultant polarity interpretation column was notably similar to that of section BJ, as will be shown in the magnetostratigraphic discussion.

5. Pole positions

The mean directions of the Carcabuey (BJ, BC) and Sierra Harana (BH) localities gave pole positions far displaced from the published data of stable Iberia. Fig. 7 shows a plot of these pole positions and Triassic through Tertiary pole positions that are clearly from the stable portion of Iberia. The published paleopoles for Iberia have been summarized by Vandenberg and Zijderveld [33]; their compilation included many paleopoles which were derived from sites located in thrust sheets that likely have rotated, and thus are not included in Fig. 7. Excluding these, only Triassic and Cretaceous paleopoles are left to estimate the position of the Iberian pole during Jurassic time. Schott et al. [34] subsequently have published a Middle Jurassic pole position from stable Iberia. However, this pole does not fall between the Triassic and Cretaceous poles; the significance of this is presently unclear. When compared to the Triassic and Cretaceous poles, the two pole positions from the Carcabuey locality are displaced clockwise by about 65°. The BH locality pole position is displaced more than 80° clockwise from the published poles. If comparison is made specifically to the Middle Jurassic paleopole [34], the amount of difference is even greater.

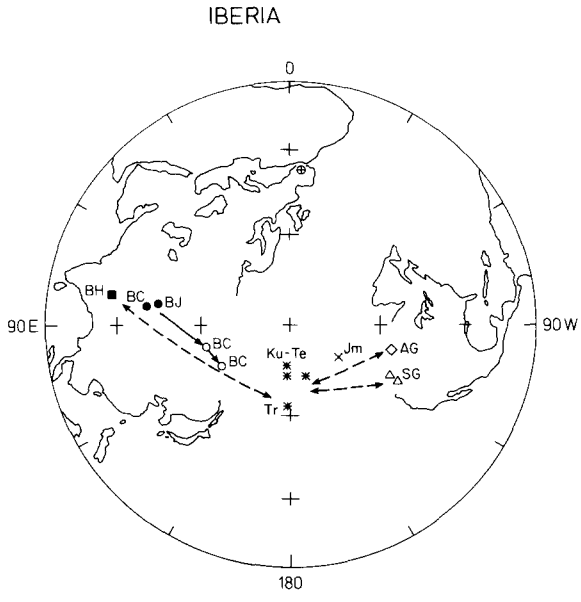


Fig. 7. Pole positions from the BJ, BC, and BH localities (solid symbols) are shown against the background of poles positions from stable Iberia. Asterisks are Triassic through Early Tertiary poles from the summary by [33]; × is the Middle Jurassic pole of [30]; BH is represented by the square, while BJ and BC are the solid circles. Also shown are the Kimmeridgian and Tithonian poles from the BC locality [1] (open circles), the Kimmeridgian and Tithonian poles from the Sierra Gorda locality of [1] (open triangles), and the Oxfordian pole of [9] (open diamond). The position of the BJ-BC locality is indicated by an encircled cross.

The Carcabuey and Sierra Harana sections have the same paleolatitude of 27° N (Table 1), suggesting that no significant north-south translation has occurred. The difference in paleolongitude suggests that a relative rotation of 20° clockwise has occurred between the two sites. Both this and the displacements with respect to paleopoles from stable Iberia, are consistent with the tectonic interpretation of the Subbetic region as being comprised of a suite of small blocks which underwent overthrusting and differential rotation during the Tertiary.

The Bajocian-Bathonian paleopoles from the BC section can be compared with poles from the Kimmeridgian and Tithonian of the same locality [1]. The significant difference observed suggests that the site underwent approximately 20° of counterclockwise rotation between the Bajocian-Bathonian and the Kimmeridgian (Fig. 7). Rotation apparently continued into the Tithonian, as the Kimmeridgian and Tithonian poles are displaced 8° from one another along the same path.

The Kimmeridgian and Tithonian pole positions from another Subbectic block to the south [1] are also shown in Fig. 7 (triangles). These are displaced clockwise about 30° with respect to the Triassic and Cretaceous paleopoles. The Oxfordian paleopole from a site in the Iberian Cordillera near Zaragoza [14] also is displaced clockwise, by nearly 40° (Fig. 7). All of these pole positions suggest that significant block rotations accompanied the thrusting deformation that affected the Iberian microplate.

6. Magnetostratigraphy

The magnetic directions at the 450 °C step and the polarity interpretation of the samples from the BJ, BC, and BH localities are displayed in stratigraphic sequence in Figs. 8, 9 and 10. The data in these figures are the same as those in Figs. 2 and 4. The ammonite zonation is shown, along with the known uncertainties in the placement of the faunal boundaries and recognized hiatuses in the sections.

A characteristic of all sections is an abundance of single samples with polarity opposite to the samples above and below them. Resampling in the BJ section above and below single-sample polarity intervals observed during the initial sampling demonstrated that many of these could be verified with additional samples and thus indicate very narrow polarity intervals, 5–10 cm thick. Thus, we tentatively associate polarity intervals with the single-sample events in all sections. However, because the possibility remains that these may represent locally remagnetized samples, such events are indicated by partial bars in the polarity interpretation diagrams. The abundance of such single-sample or other narrow polarity intervals indicates that the actual polarity pattern for the Bajocian-Bathonian is very likely to be much more complex than that displayed by any single section.

The polarity patterns of the sections are compared in Fig. 11. In order to compare sequences that have such a rapid reversal frequency, the exact positions of the samples and the location of unsampled portions of the sections must be considered. The locations of hiatuses also become very important.

The two Carcabuey sections, BJ and BC, have fairly similar magnetostratigraphic patterns. The polarity around the *T. (R.) sofanus/Z. zigzag* boundary (Fig. 11) is nearly identical. However,

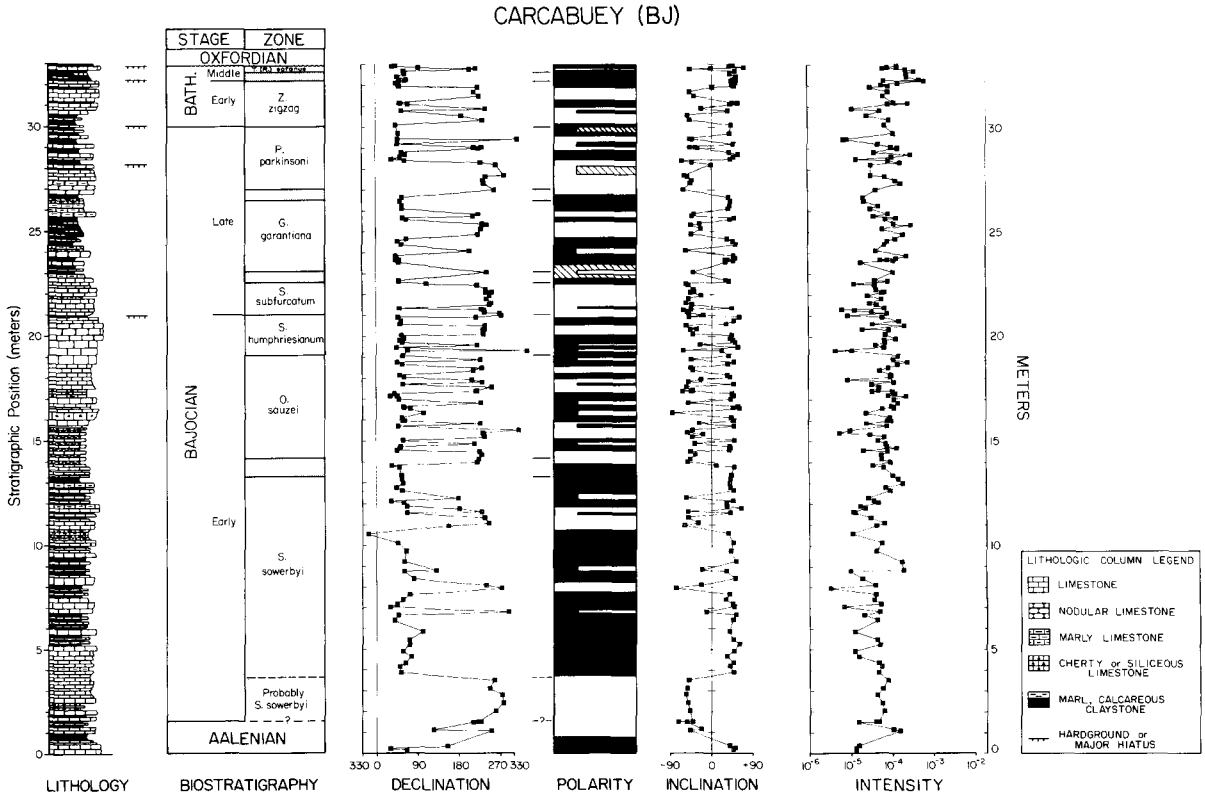


Fig. 8. Stratigraphic distribution of paleomagnetic directions at the BJ locality after thermal demagnetization. Stippled pattern indicates known uncertainties in the placement of ammonite zones. Diagonal pattern represents areas in which sampling is inadequate. Single sample polarity intervals are indicated by partial bars. Black (white) is normal (reversed) polarity; intensity is in units of A/m.

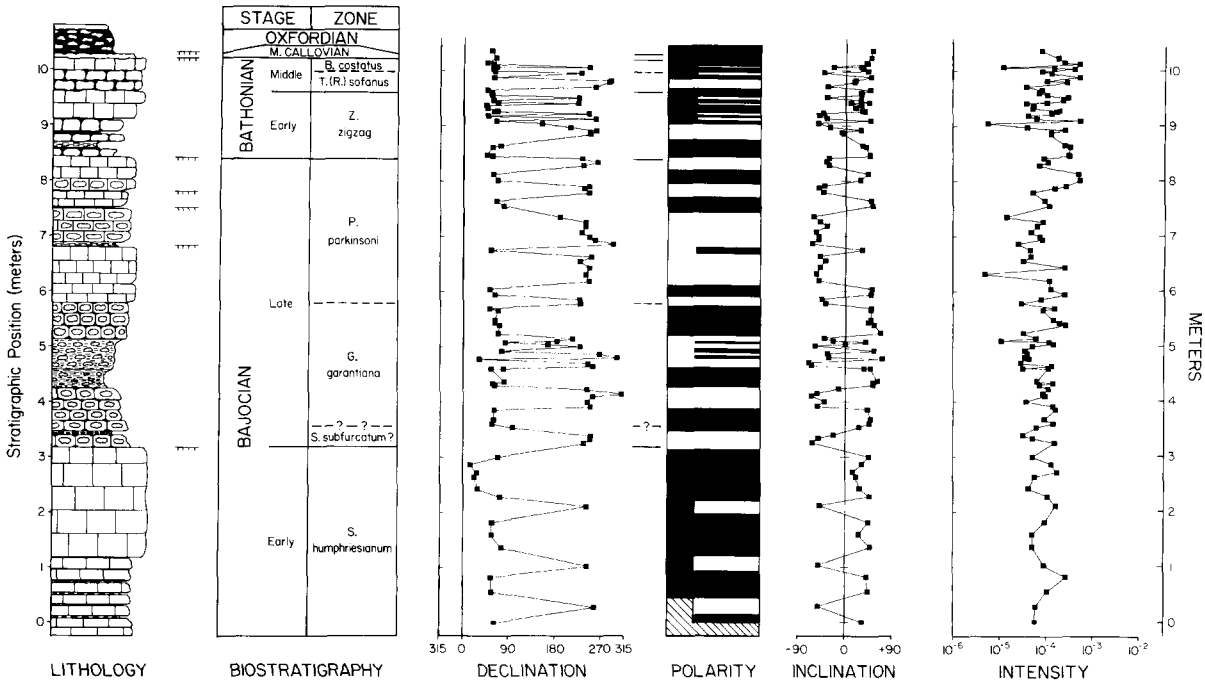


Fig. 9. Stratigraphic distribution of paleomagnetic directions at the BC locality after thermal demagnetization. Symbols as in Fig. 8.

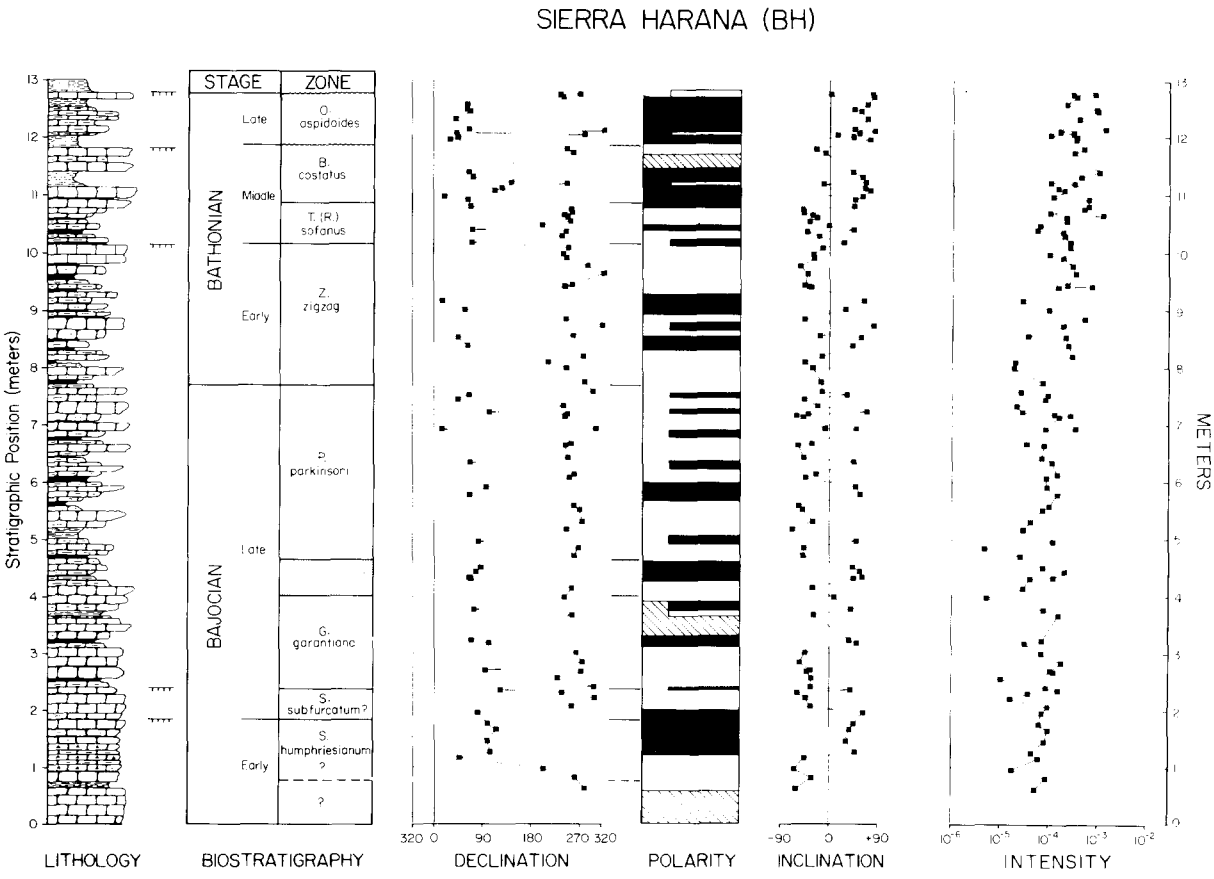


Fig. 10. Stratigraphic distribution of paleomagnetic directions at locality BH after thermal demagnetization. Symbols as in Fig. 8.

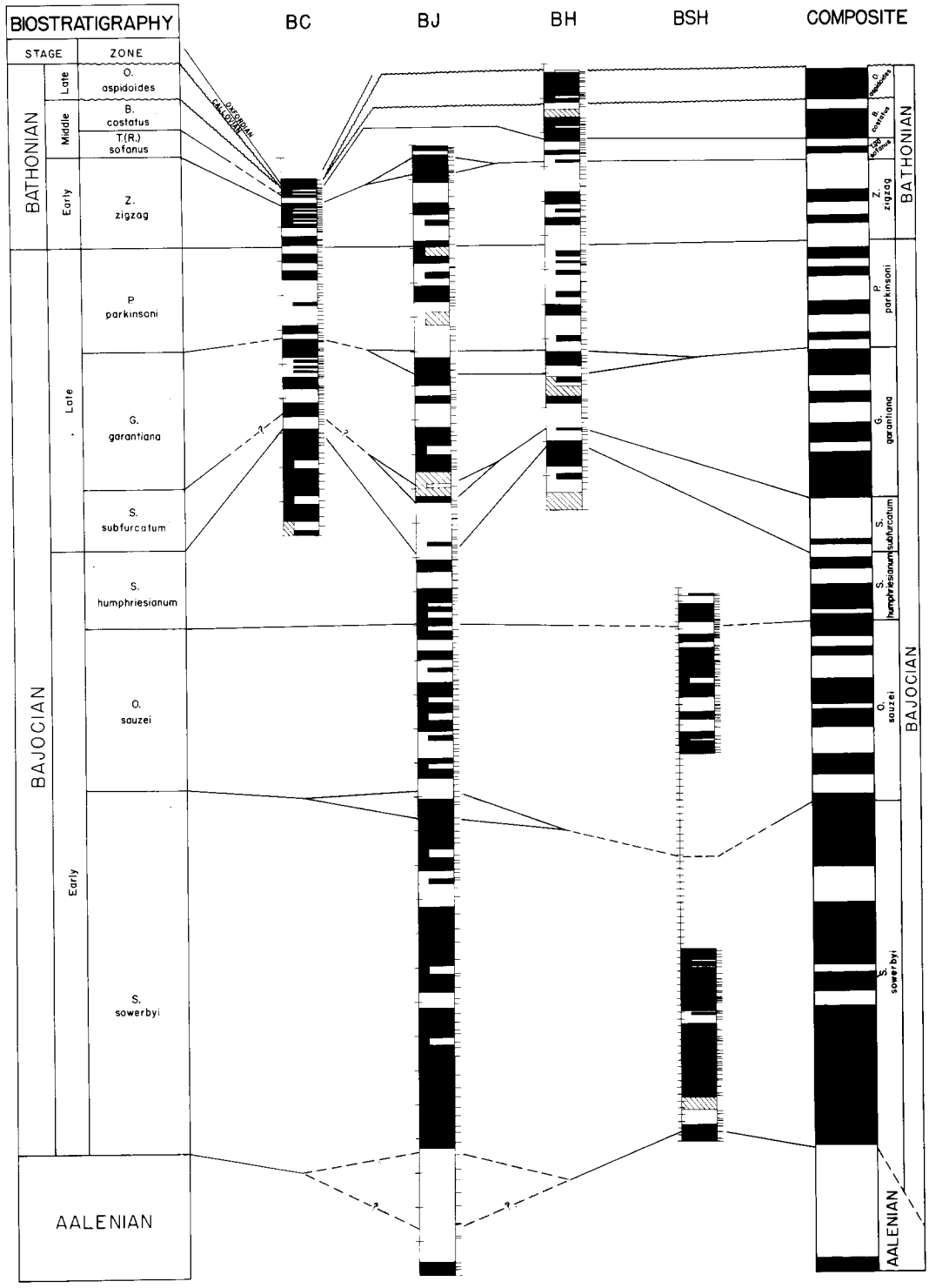


Fig. 11. Correlation of the magnetostratigraphic interpretations for the localities. Diagonal pattern indicates areas in which sampling is inadequate; stippled pattern indicates age uncertainty. Sample positions are indicated at the right of each column. A schematic generalized column is shown on the far right. This column is not intended to depict the exact geomagnetic polarity history of this time, but a representation of the general features of the field as indicated by the combined records.

the *Z. Zigzag* Zone appears to be thicker at the BJ locality than at BC. A thinning of section at BC relative to BJ, in fact, is indicated by all ammonite zones. The magnetic pattern within the *P. Parkinsoni* Zone is similar within the two sections, and the *G. Garantiana* Zone also shows reasonably comparable patterns. Within BC, the *S. Subfurcatum* Zone cannot reliably be identified, and its apparent absence suggests a major hiatus in section BC. A hiatus at this point also would explain the apparent mismatch of polarity patterns at and below this level; the assumption of a hiatus makes the polarity patterns of the two sections correspond relatively well.

The polarity sequence of the BH section is generally the same as that of BJ/BC (Fig. 11). The lower *Z. Zigzag* and the *P. Parkinsoni* zones of BH and BJ/BC are similar, however, considerable difference exists in the upper *Z. Zigzag* and *T. sofanus* zones. Below this, the upper part of the *G. Garantiana* Zone is fairly similar in both localities, but a hiatus probably exists in the BH section within the lower *G. Garantiana* and the *S. Subfurcatum* zones. The fact that the *G. Garantiana* Zone is represented by single, unusually thick and extensively nodular limestone bed, and the mismatch of polarity, both suggest a hiatus. Below it, the *S. Humphriesianum* Zone (the lowest zone exposed at BH) is only tentatively identified, and thus cannot be compared to other sections. Thus, allowing for two significant hiatuses, and with the exception of the discrepancy around the Early/Middle Bathonian boundary, the lower Bathonian and upper Bajocian magnetostratigraphic sections of BH, BJ, and BC display a broadly similar magnetic polarity pattern, which is one of quite frequent reversals.

For the lower Bajocian, the polarity sequence of BJ only can be compared to the tentative polarity sequence of BSH. Only the upper and lower portions of the BSH section were sampled, and the tentative nature of this polarity interpretation was stressed earlier. Although the polarity structure of BSH is tenative, it is notably similar to the polarity sequence exhibited by the BJ section (Fig. 11). This similarity suggests that the polarity pattern of BJ is likely to be a reasonable representation of the geomagnetic polarity for this time interval.

7. Discussion and conclusions

The polarity sequences obtained from these four well-dated stratigraphic sections of Early Bajocian through early Late Bathonian age indicate frequent reversals of the geomagnetic field. All sequences indicate frequent reversals, and display similar polarity sequences for the same time intervals. The details of the patterns differ and all sections have an abundance of single-sample polarity intervals. Remagnetization, perhaps during diagenesis, is one explanation for these features. However, the lack of detailed correlation and the single-sample polarity intervals were undoubtedly generated, at least in part, by the conditions under which these sequences were deposited. These were slowly deposited sedimentary records, in which relatively short hiatuses are common, and they were deposited during a time of relatively high reversal frequency of the geomagnetic field. These two circumstances predict the type of records and lack of correlation obtained, and is not necessary to invoke remagnetization, although it certainly cannot be excluded.

The frequency of reversals indicated by these sequences is high in comparison to the general history of the geomagnetic field as we currently know it. Using estimates [30,31] of the duration of the Bajocian and Bathonian stages, an approximate reversal frequency was obtained. The uncertainties of the absolute time scale for the Jurassic are still large and the *exact* polarity pattern is not known, hence these numbers can only provide a basis for a very general comparison to other time periods. Use of the timescale of [2] gives the same result. The composite illustrated in Fig. 11 has polarity intervals of an average duration of about 260,000 years. This composite purposely is weighted against the many single-sample reversals; an average duration of 170,000–180,000 years is indicated if the single sample intervals are considered. Because of this and the circumstances of rapid reversals recorded in slow-sedimentation sequences, the true reversal frequency probably is greater than one every 260,000 years. Nevertheless, in comparison to the polarity history of the geomagnetic field, 260,000 years is at the high end of the known frequency range. Other time periods with such frequencies are the Pliocene-Miocene, which has an average duration of about 225,000

years per polarity interval [2,35], the Early Cretaceous with an average of approximately 400,000 years [2,35], and the Oxfordian-Callovian, with an average duration of about 215,000 years [10,31,13]. Magnetostratigraphic study of the Oxfordian suggests an even more rapid rate, very approximately 130,000 years [14,31].

High reversal frequencies appear to be a characteristic of the Jurassic, particularly of the Jurassic older than the Kimmeridgian, and older than marine magnetic anomaly M25. High reversal frequency in the Oxfordian [10–14] continues through the Callovian [15,12,13]. As shown by this study and others [18–22], the Bathonian and Bajocian continue the rapid reversal rate. The Jurassic reversal frequency as it is known thus far appears to be marginally faster in the Oxfordian-Callovian. High frequency of reversals probably contributes towards the "quietness" of ocean crust of the Late/Middle Jurassic. The Jurassic quiet zones are probably Late Bathonian to Oxfordian age. Rapidity of reversals is observed before the start of the quiet zones (this study) and during the time of the late stages of the quiet zones [14]. Frequent reversals would result in closely spaced, narrow magnetic anomalies. Considering the considerable depth of Middle/Late Jurassic crust, such narrow anomalies would be difficult to resolve at the sea surface. The frequency of reversals as observed in well-dated magnetostratigraphic sequences [1,14] appears to be decreasing from the Oxfordian to the Tithonian, which may contribute to the gradually more resolvable nature and higher amplitude of the observed magnetic anomalies.

In conclusion, four sequences of well-dated pelagic limestone indicate that the Bajocian-Bathonian was a time of rapid reversals of the geomagnetic field. The frequency is on a par with the highest frequencies yet observed for the earth's magnetic field. Because of the reversal frequency and the slow sedimentation rate, the composite of Fig. 11 probably is not the exact polarity pattern for this time, and probably is an underestimate of the actual number of reversals. It may not be possible to obtain a complete record of the geomagnetic field during this time, unless continuous, rapidly deposited stratigraphic sections can be studied. However, the rapidity of reversals is distinctive, and may serve as a useful stratigraphic correlation tool, in a manner somewhat analogous to the Late Paleozoic (Kiaman) long reversed polarity interval or the Cretaceous long normal interval.

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